



KTU NOTES

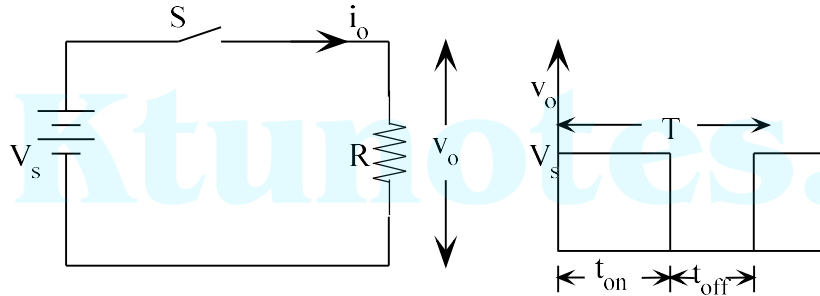
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PRINCIPLE OF OPERATION OF CHOPPER

A chopper is simply an ON-OFF switch that connects and disconnects a load at a very fast rate. When the switch S is closed, the supply voltage V_s appears across the load. When it is open, the load is disconnected from the supply. The average dc output voltage is obtained by controlling the switch ON period t_{on} and switch OFF period t_{off} .



The output voltage v_o is a chopped voltage derived from the supply voltage V_s . The average output voltage is obtained by controlling the switch ON period t_{on} and switch OFF period t_{off} .

$$\text{Average output voltage, } V_o = \frac{1}{T} \int_0^{t_{on}} v_o dt = \frac{t_{on}}{T} V_s = D V_s$$

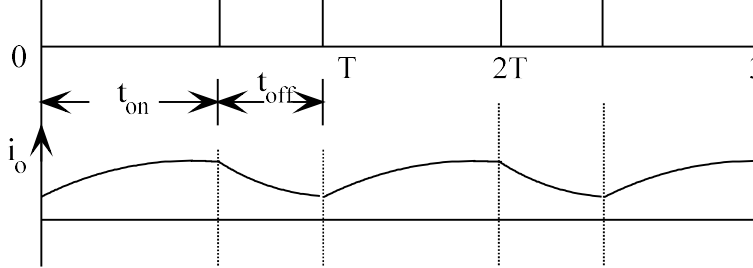
where t_{on} is the on-time of the chopper,

T is the chopping period ; $f_s = \frac{1}{T}$ = switching frequency

D is the duty ratio of the chopper; $D = \frac{t_{on}}{T}$; D varies from 0 to 1

STEP DOWN CHOPPER

When the switch S is turned on at $t=0$, the supply is connected to the load and the current i_o builds up exponentially. When the switch is turned off at $t = t_{on}$, the load current i_o decays exponentially through the load inductor and $v_o = 0$. Energy stored in the load inductance is gradually dissipated in the load resistor. At $t = T$, switch S is turned on again and the cycle repeats. The waveforms of the load voltage v_o and load current i_o are shown below. It is assumed that the load current i_o is continuous – never reaches zero during off period (switching frequency is high).

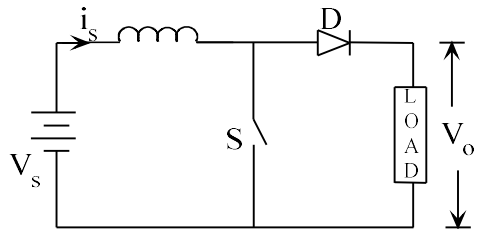


Average voltage, $V_o = \frac{1}{T} \int_0^{t_{on}} v_o dt = \frac{t_{on}}{T} V_s = DV_s$

The output voltage varies linearly with the duty ratio of the chopper. The controlled in the range $0 < V_o < V_s$. This configuration of the chopper is known as a step-down chopper. If the switch S is a GTO, a positive gate pulse will turn it on and a negative gate pulse will turn it off. If the switch is a transistor, the base current will control the on and off periods. If the switch is an SCR, a commutation circuit is required to turn it off.

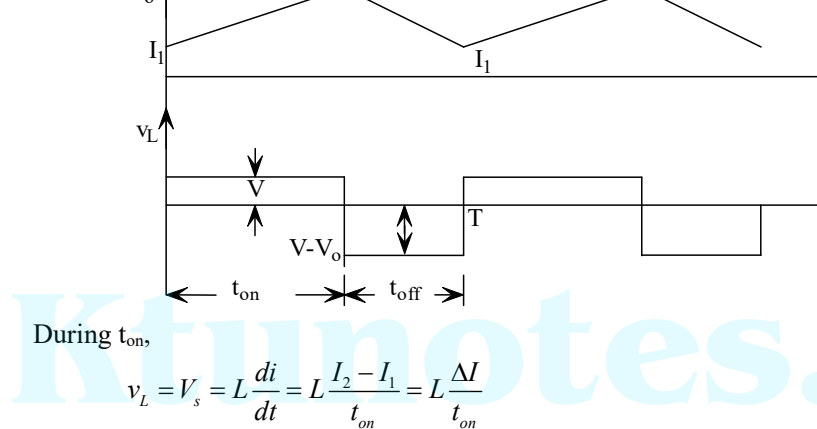
STEP-UP CHOPPER

This chopper configuration produces an average output voltage more than the supply voltage (i.e., $V_o > V_s$).



Assumption : Output voltage is assumed to be continuous and ripple free.

When the chopper is on, the inductor L is connected to the supply voltage V_s . Inductance gets charged. Diode D is reverse biased. Load is connected to the source. The inductor current i_L increases linearly ($v_L = L \frac{di_L}{dt} = +V_s$ or $\frac{di_L}{dt} = \frac{V_s}{L}$, i_L is increasing linearly).



$$\text{or } \Delta I = V_s \frac{t_{on}}{L} \quad \text{----- (1)}$$

During t_{off} ,

$$v_L = V_s - V_o = L \frac{I_1 - I_2}{t_{off}} = -L \frac{\Delta I}{t_{off}}$$

$$\text{or } \Delta I = -(V_s - V_o) \frac{t_{off}}{L} \quad \text{----- (2)}$$

Comparing equations (1) and (2),

$$V t_{on} = (V_o - V_s) t_{off}$$

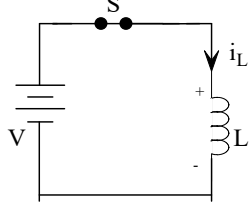
$$V_o = V_s \frac{t_{on} + t_{off}}{t_{off}} = \frac{T}{T - t_{on}} V_s$$

$$V_o = \frac{1}{1 - D} V_s$$

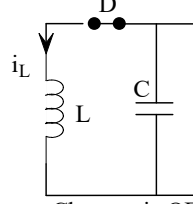
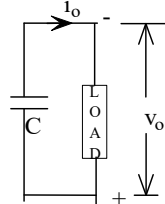
By varying D from 0 to 1, output voltage V_o can be varied from V to infinity. V_o is greater than the input voltage V_s ; hence it is a step up chopper.

Note : Average voltage across inductance is zero under steady state condition. $V t_{on} = (V_o - V_s) t_{off}$ (Positive & negative areas of v_L are equal)

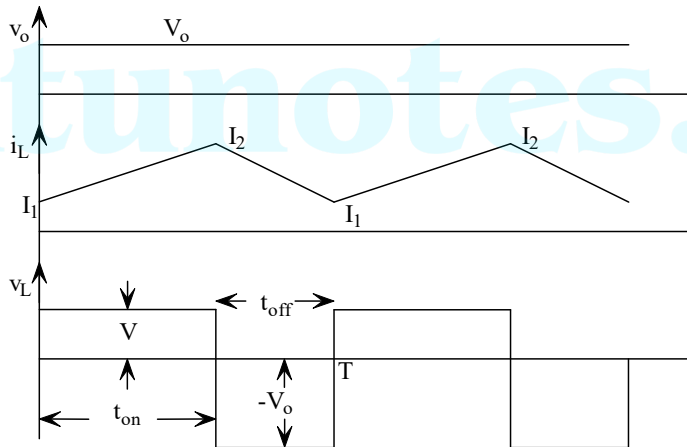
Note: This principle of operation is utilized in the regenerative braking of a dc motor. In this case, the armature voltage of the dc machine and V_o represents the dc supply, power is fed back to the supply by the decreasing motor voltage V to the fixed supply voltage V_o by proper adjustment of D .



Chopper is ON



Chopper is OFF



During t_{on} ,

$$v_L = V = L \frac{di}{dt} = L \frac{I_2 - I_1}{t_{on}} = L \frac{\Delta I}{t_{on}}$$

$$\text{or } \Delta I = V \frac{t_{on}}{L} \quad \text{----- (1)}$$

During t_{off} ,

$$v_L = -V_o = -L \frac{I_2 - I_1}{t_{off}} = -L \frac{\Delta I}{t_{off}}$$

$$\text{or } \Delta I = V_o \frac{t_{off}}{L} \quad \text{----- (2)}$$

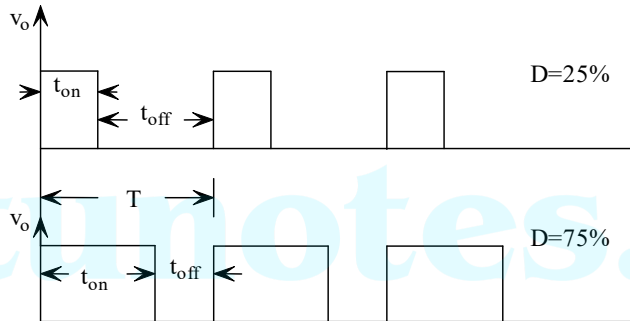
From equations (1) and (2),

$$V t_{on} = V_o t_{off}$$

$$V_o = V \frac{t_{on}}{t_{off}} = \frac{t_{on}}{T - t_{on}} V$$

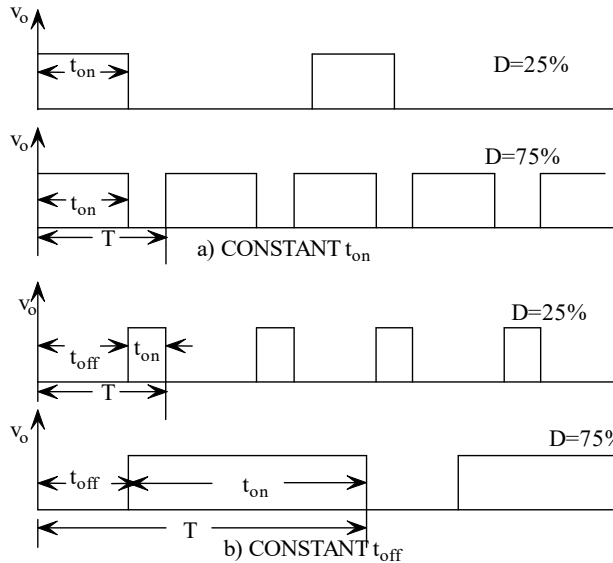
switching frequency constant)

In figure (a), $t_{on} = \frac{T}{4}$, $D = 25\%$. In figure (b), $t_{on} = \frac{3T}{4}$, $D = 75\%$.

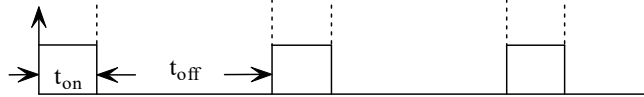


b) Variable Frequency System (Frequency Modulation Control) : t_{on} (or t_{off}) is varied.

In figure (a), t_{on} is kept constant, T is varied. In figure (b), t_{off} is kept constant, T is varied.



Note: Constant frequency system (PWM) is the preferred scheme for chopper drive.

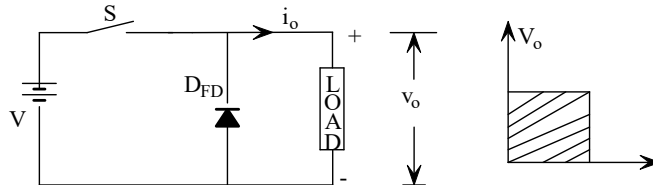


Regenerative Braking of DC motors

In regenerating braking, generated energy is supplied to the load. This occurs when the back EMF $E > V$ and I is negative. For a source of fixed voltage (of rated value), regeneration is possible only for speeds higher than rated. When the machine regenerates, its speed is higher than the rated speed. Consequently, the regenerated power flows into the loads connected to the supply. The supply is relieved from supplying this much amount of power. The regenerative braking is possible only when there are loads connected to the line and they are in need of power. If there is no load, the regenerated power will not be absorbed by the loads. Hence, regenerative braking is possible only when there are enough loads to absorb the regenerated power. When the regenerated energy can be stored in the battery.

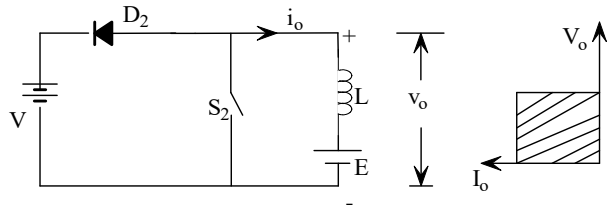
TYPES OF CHOPPERS

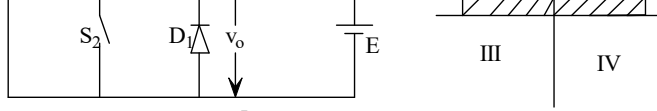
First Quadrant Chopper or Type A Chopper



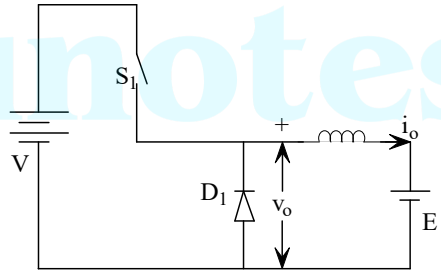
Average values of both load voltage and current, i.e. V_o and I_o are always positive. The output voltage V_o is always from source to load. This chopper is also called step-down chopper as V_o is always less than input dc voltage V .

Second Quadrant Chopper or Type B Chopper



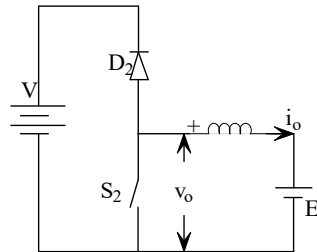


If the chopper S_1 and the diode D_1 are operated, the system operates as a step-down chopper. The output voltage V_o is either V (when S_1 is on) or zero (when S_1 is off). The average value of the output voltage is positive and the output current i_o flows in the positive direction. The chopper, therefore, operates in the first quadrant.



First Quadrant Operation

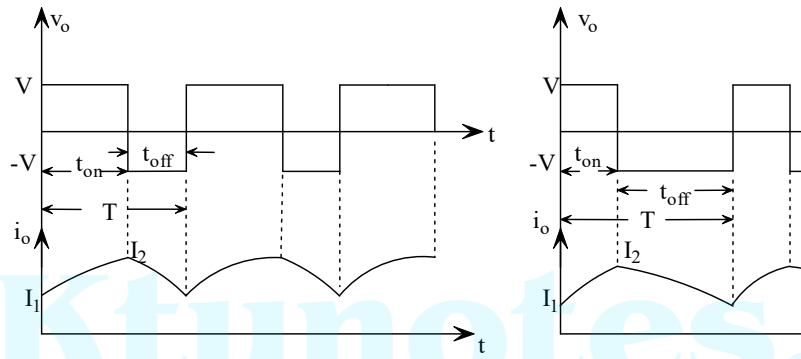
If, however, the chopper S_2 and the diode D_2 are operated, the system operates as a step-up chopper with E as source. (if the load is a dc machine, it operates in the regenerative braking mode). The output voltage V_o is either zero (when S_2 is on) or V (when S_2 is off). The average value of the output voltage is positive, but the output current now flows in the negative direction. The chopper then operates in the second quadrant.



Second Quadrant operation

Note : S_1 and S_2 should not be operated simultaneously which leads to short circuit of the source.

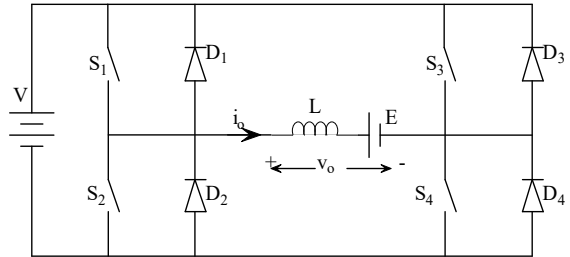
Load current always flows in positive direction.



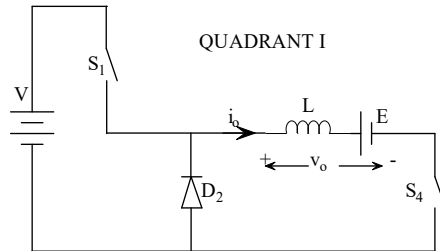
Average output voltage, $V_o = V \frac{t_{on} - t_{off}}{T}$ and duty ratio, $D = \frac{t_{on}}{T}$

If D is 0.5, $V_o=0$; if D is greater than 0.5, V_o is positive and if D is less than 0.5, V_o is negative.

Four Quadrant Type E Chopper

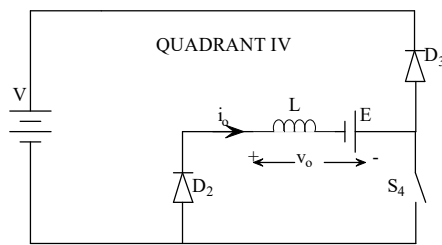


In this chopper, both the load voltage and load current may be either positive or negative. S_1 and D_3 constitute one type C chopper and S_2 , S_3 , D_1 and D_4 constitute another type C chopper.



QUADRANT I

S_2 & S_4 are kept OFF;
 S_1 is kept ON;
 S_1 is operated ON & OFF
 V_o and I_o are positive



E is reversed;
 S_1, S_2 & S_3 are
 S_4 is operated ON
 V_o is negative

NOTE : For Quadrant 1, S_1 is operated ON & OFF; for Quadrant 2, S_2 is operated ON & OFF; for Quadrant 3, S_3 is operated ON & OFF; for Quadrant 4, S_4 is operated ON & OFF.

Problem : A chopper circuit is operating on TRC principle at a frequency of 1 kHz. If the load voltage is 170V, compute the conduction and blocking period.

$$T = 1/f = 0.5 \text{ ms}$$

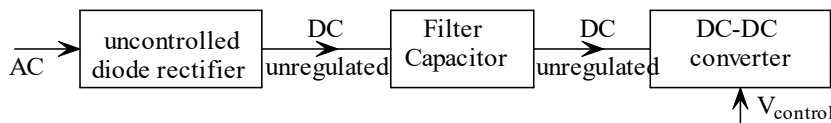
$$t_{on} = T \frac{V_o}{V} = 0.386 \text{ ms}$$

$$t_{off} = T - t_{on} = 0.114 \text{ ms}$$

SWITCHING REGULATORS

Unregulated dc is converted to regulated dc at a desired voltage level.

Here, self-commutated devices like power BJT, power MOSFET, IGBT and thyristors. It is used for low and medium power applications whereas thyristors are used for high power applications.



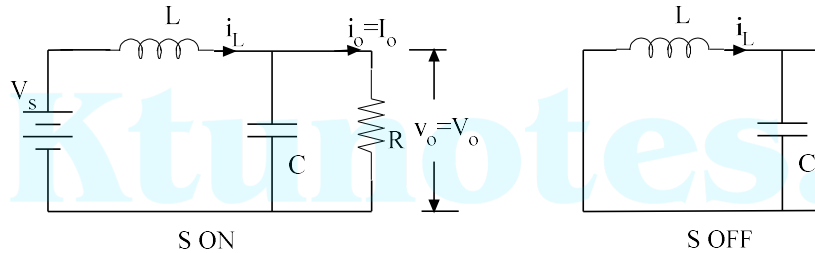
Different types of switching regulators are i) Buck regulator ii) Boost regulator iii) Buck-Boost regulator and iv) Cuk regulator.

DC-DC power converters are widely used in regulated switch-mode dc motor drive applications. The input to the DC-DC converter is an unregulated dc voltage. The DC-DC converter produces a regulated output voltage V_o , having a magnitude that differs from the input voltage.

Different DC-DC converters are

1. Step-down (buck) converter

energy to the load as well as to the inductor. During the interval when the switch is ON, current flows through the diode, transferring some of its stored energy to the load. The filter capacitor at the output is assumed to be very large so that $v_o(t) = V_o$. For a buck converter, average inductor current I_L is equal to the average inductor current. The average capacitor current in steady state is zero.



Note:

- 1) The principle of inductor volt-second balance states that the average voltage applied across an ideal inductor winding must be zero.
- 2) The principle of capacitor amp-second or charge balance, states that the average current flowing through an ideal capacitor must be zero.

Continuous conduction mode (CCM) and discontinuous conduction mode (DCM) are defined with respect to the inductor current. If the current through the inductor is continuous and never reaches zero at any instant in a switching period, then the converter operation is referred to as CCM. If the current through the inductor becomes zero for some portion of the switching period, then the converter operation is referred to as DCM.

CONTINUOUS CONDUCTION MODE

(Inductor current flows continuously $i_L(t) > 0$)

Average voltage across inductance over once cycle is zero. Hence,

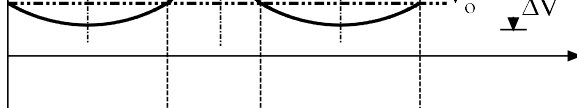
$$(V_s - V_o) \times DT - V_o \times (1 - D)T = 0$$

Average output voltage, $V_o = V_s \times D$

Assuming ideal circuit,

$$V_s I_s = V_o I_o$$

$$I_o = \frac{V_s I_s}{V_o} = \frac{V_s I_s}{D V_s} = \frac{I_s}{D}$$



Buck converter is equivalent to a step-down transformer with turns ratio D .

EXPRESSION FOR PEAK-TO-PEAK RIPPLE CURRENT

During S ON,

Voltage across inductance, $V_s - V_o = L \frac{I_2 - I_1}{DT} = L \frac{\Delta I}{DT}$

$$\Delta I = \frac{DT \times (V_s - V_o)}{L} = \frac{V_s D(1-D)}{fL} \text{ where } T = \frac{1}{f} \text{ and } V_o = V_s D$$

EXPRESSION FOR PEAK TO PEAK RIPPLE VOLTAGE

With very large value of C , $v_o = V_o$

But, for practical value of C , output voltage contains ripple.

Instantaneous value of inductor current, $i_L = i_c + i_o$

Taking average over one cycle,

$$I_L = I_o \quad (\text{Since average value of capacitance current is zero})$$

Note: Load current is assumed to be ripple-free while deriving the expression for

$$\Delta V = \frac{\Delta Q}{C} = \frac{1}{C} \times \left(\frac{1}{2} \times \frac{\Delta I}{2} \times \frac{T}{2} \right) = \frac{1}{C} \times \left(\frac{1}{2} \times \frac{V_s D(1-D)}{2fL} \times \frac{1}{2f} \right)$$

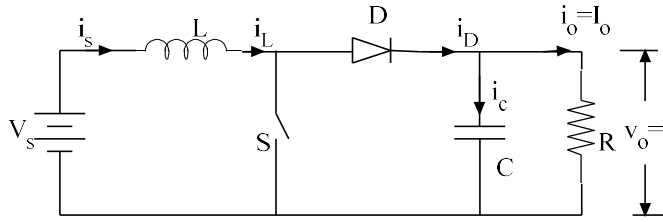
$$\Delta V = \frac{V_s D(1-D)}{8f^2 LC}$$

Problem 1:- A buck converter topology was used to obtain the required dc load voltage from a dc input of 12V. The peak to peak output ripple was restricted to 1% of the output voltage. The switching frequency is 25kHz. For maintaining the peak to peak ripple current of inductor less than 10% of the average load current, determine i) the duty cycle ii) the filter inductance L and capacitance C .

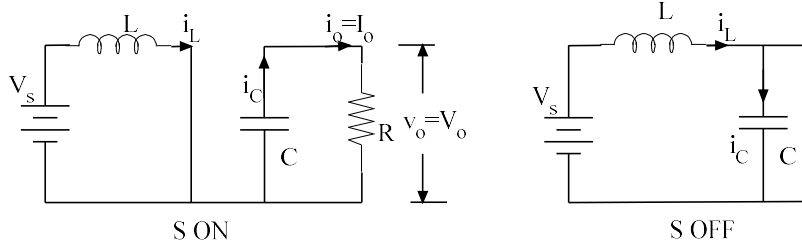
(Ans: $D=0.417$; $L=145.83\mu\text{H}$; $C=200\mu\text{F}$)

Problem 2:- A buck converter has a switching frequency of 10kHz. It is used to obtain an output voltage of 48V across 1Ω resistor when fed from a 192V dc source. The filter inductance is $200\mu\text{H}$. Determine the duty ratio, average load current, average

In Boost converters, the output voltage is always greater than the input voltage.



When the switch is ON, the diode is reverse biased, thus isolating the output from the input. The inductor supplies energy to the inductor. When the switch is OFF, the output stage receives energy from the inductor as well as from the input. The output filter capacitor is assumed to be large enough to maintain a constant output voltage $v_o(t) = V_o$.



CONTINUOUS CONDUCTION MODE (Inductor current flows continuously)

Average voltage across inductance over once cycle is zero. Hence,

$$V_s \times DT + (V_s - V_o) \times (1 - D)T = 0$$

$$\text{Average output voltage, } V_o = \frac{V_s}{1 - D}$$

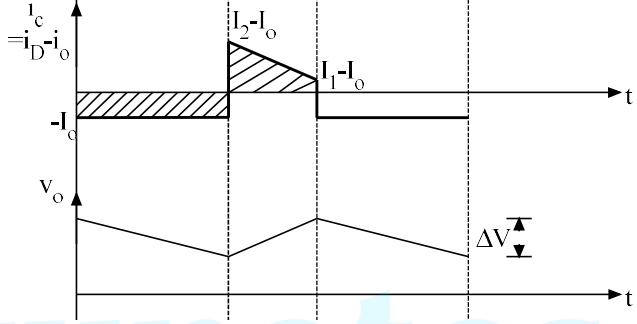
$$D=0 \rightarrow V_o=V_s \quad \& \quad D=1 \rightarrow V_o=\infty$$

Assuming ideal circuit,

$$V_s I_s = V_o I_o$$

$$I_o = \frac{V_s I_s}{V_o} = (1 - D) I_s$$

Boost converter is equivalent to a step-up transformer with turns ratio D .



EXPRESSION FOR PEAK-TO-PEAK RIPPLE CURRENT

During S ON,

$$\text{Voltage across inductance, } V_s = L \frac{I_2 - I_1}{DT} = L \frac{\Delta I}{DT}$$

$$\Delta I = \frac{DT \times V_s}{L} = \frac{V_s D}{fL} \text{ where } T = \frac{1}{f} \text{ and } V_o = \frac{V_s}{1-D}$$

EXPRESSION FOR PEAK TO PEAK RIPPLE VOLTAGE

With very large value of C, $v_o = V_o$

But, for practical value of C, output voltage contains ripple.

Note: Load current is assumed to be ripple-free while deriving the expression for

$$\Delta V = v_c(0) - v_c(t_{on}) = \frac{1}{C} \int_0^{t_{on}} i_c dt = \frac{1}{C} \int_0^{t_{on}} I_o dt = \frac{I_o DT}{C}$$

$$\Delta V = \frac{I_o D}{fC}$$

Problem 3: In a boost converter, the duty ratio D is adjusted to regulate the output voltage. If the input voltage varies in a wide range of $12V \leq V_s \leq 36V$. The maximum power is 120W. For stability reasons, the input current is discontinuous. The switching frequency is 20kHz. Assuming ideal devices and components and filter capacitor of very large value, calculate the required inductor L.

Ans:-

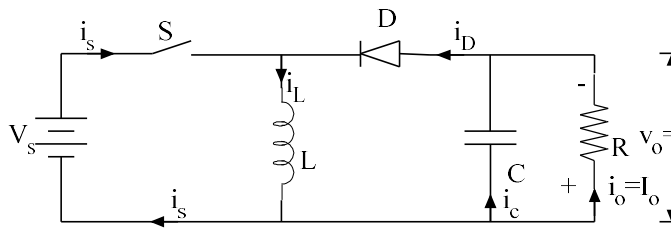
$$V_s = 12V \rightarrow D = 0.75$$

$$V_s = 36V \rightarrow D = 0.25$$

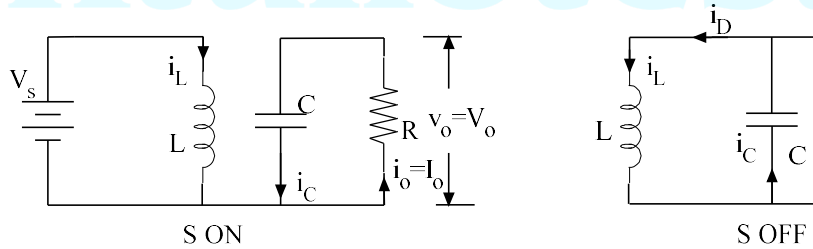
Assume i_L is at the boundary of continuous and discontinuous conduction.

$$\text{Maximum value of output current, } I_{oB} = 120/48 = 2.5A$$

$$R_{CR} = 48/2.5 = 19.2\Omega$$



When the switch is ON, the diode is reverse biased and input provides energy to the inductor. When the switch is OFF, the energy stored in the inductor is transferred to the output. The output filter capacitor is assumed to be large enough to maintain a constant output voltage $v_o(t) = V_o$.



CONTINUOUS CONDUCTION MODE (Inductor current flows continuously)

Average voltage across inductance over once cycle is zero. Hence,
 $V_s \times DT - V_o \times (1-D)T = 0$

Average output voltage, $V_o = V_s \frac{D}{1-D}$

$D=0 \rightarrow V_o=0$

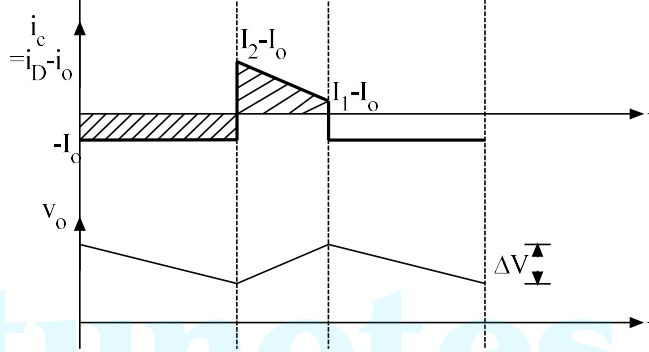
$D=0.5 \rightarrow V_o=V_s$

$D=1 \rightarrow V_o = \infty$

Assuming ideal circuit,

$$V_s I_s = V_o I_o$$

$$I_o = \frac{V_s I_s}{V_o} = \frac{(1-D)I_s}{D}$$



EXPRESSION FOR PEAK-TO-PEAK RIPPLE CURRENT

During S ON,

$$\text{Voltage across inductance, } V_s = L \frac{I_2 - I_1}{DT} = L \frac{\Delta I}{DT}$$

$$\Delta I = \frac{DT \times V_s}{L} = \frac{V_s D}{fL} \text{ where } T = \frac{1}{f} \text{ and } V_o = \frac{DV_s}{1-D}$$

EXPRESSION FOR PEAK TO PEAK RIPPLE VOLTAGE

With very large value of C, $v_o = V_o$

But, for practical value of C, output voltage contains ripple.

Note: Load current is assumed to be ripple-free while deriving the expression for

$$\Delta V = v_c(0) - v_c(t_{on}) = \frac{1}{C} \int_0^{t_{on}} i_c dt = \frac{1}{C} \int_0^{t_{on}} I_o dt = \frac{I_o DT}{C}$$

$$\Delta V = \frac{I_o D}{fC}$$

AVERAGE VALUE OF INDUCTOR CURRENT

$$i_L = i_s + i_c + i_o$$

Under steady state, taking average values

$$I_L = I_s + I_o \text{ (Average value of capacitor current is zero)}$$

$$I_L = I_s + I_o = \frac{DI_o}{1-D} + I_o = \frac{I_o}{1-D}$$

$$I_L = \frac{I_o}{1-D}$$

Problem 5 : The buck-boost converter is used to supply an average load current. The converter is fed from a 12V source and is operating at a frequency of 25kHz.

